

Canon as Sphere-Packing Lattice

First-Principles Construction of the OMI Canon Spine

0. Purpose

This document defines the 24-file OMI canon spine as a bounded construction lattice.

The point is not to claim that OMI literally is a Bravais lattice, the Leech lattice, or the Lorentzian lattice

`II25,1`.

The point is to use those structures as disciplined mental models for balance, order, primitive cells, symmetry, reflection, and bounded construction.

OMI needs a way to maintain order while allowing derived declarations to grow.

The canon spine supplies that order.

The 24 root documents form the primitive canonical packing.

Derived declarations may extend, refine, quote, project, or specialize the canon, but they do not create new authority.

They must fit inside the bounded relation-space defined by the canon.

1. First Principle: A Canon Is Not a List

A list is only an ordering.

A canon is a bounded construction space.

A list says:

this comes before that

A canon says:

this relation may be constructed here,
under these boundaries,

with these neighboring constraints,
and with this validation path.

Therefore the OMI canon spine should not be understood as merely:

00, 01, 02, ..., 24

It should be understood as a packed configuration of doctrinal cells.

Each root document is a boundary cell.

Each boundary cell has neighbors.

Each neighbor constrains interpretation.

Each interpretation constrains derived declaration.

The canon is therefore closer to a lattice or sphere packing than a table of contents.

2. Bravais Lattice Mental Model

A Bravais lattice begins from primitive translation vectors.

A point in the lattice is generated by integer combinations of those vectors:

$$\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3$$

The important idea is not the physical crystal.

The important idea is this:

primitive vectors generate repeatable order

A lattice does not need every point to be separately invented.

Once the primitive basis is fixed, the space can be generated lawfully.

OMI uses the same design intuition.

The canon spine supplies primitive doctrinal vectors.

A derived declaration is valid only when it is generated from those primitive vectors without breaking the lattice.

In OMI terms:

```
primitive vectors = canonical documents  
unit cell = local relation basis  
lattice point = derived declaration  
translation = lawful projection, quotation, propagation, or carrier  
transformation
```

A derived declaration must not float freely.

It must be placed.

It must be addressable.

It must preserve relation identity.

It must remain reducible to the canonical basis.

3. Unit Cell Mental Model

A primitive unit cell is the smallest repeating unit that can reproduce the lattice.

In OMI, the local primitive cell is not a cube.

It is the tetrahedral sphere basis of the Omi-Ring.

The tetrahedral basis gives the smallest local relation structure:

```
point  
edge  
face  
volume
```

or, in OMI language:

```
root  
relation  
witness  
closure
```

This basis is local.

The 24-file canon is global.

The relation between them is:

```
Omi-Ring tetrahedral basis = local primitive cell
24-file canon spine       = global sphere-packing boundary
```

Every derived declaration must be able to circumscribe the tetrahedral basis without tearing it away from the global canon.

That means every declaration must answer:

```
Where is its root?
What relation does it quote?
What witness can inspect it?
What closure carries it forward?
What validation path adjudicates it?
```

4. N-Ball and N-Sphere Interpretation

An n-ball is the bounded interior.

An n-sphere is the boundary surface.

For OMI:

```
n-ball   = allowed construction interior
n-sphere = canonical boundary of interpretation
```

The 24-file canon spine is the n-sphere boundary.

It defines what the construction space is allowed to touch.

Derived declarations live in the n-ball interior.

They may move, refine, and compose inside the boundary, but they may not redefine the boundary.

Therefore:

```
canon files = boundary spheres
derived declarations = interior construction balls
validation = test of lawful placement
projection = visible surface after acceptance
```

A derived declaration that cannot be placed inside the bounded canon is not accepted OMI state.

It may remain a candidate.

It may remain a note.

It may remain an external carrier artifact.

But it is not canonically accepted state.

5. Sphere Packing as Canon Discipline

Sphere packing asks how equal or structured spheres can be arranged without overlap while preserving density, symmetry, and contact.

OMI uses this as a mental model for doctrinal balance.

Each canonical document is a sphere-cell.

It must touch neighboring cells without collapsing into them.

For example:

```
OMI-LISP touches BLACKBOARD,
but OMI-LISP is not the BLACKBOARD.
```

```
BLACKBOARD touches NOTATION-MULTIPLEXING,
but BLACKBOARD is not the carrier.
```

```
PROJECTION-FACES touch TRANSFORM-CARRIERS,
but projection is not authority.
```

```
CANONICAL-CONSTITUTION touches MASTER-ARCHITECTURE,
but architecture is not law.
```

The packing rule is:

touching is allowed
overlap of authority is not allowed

This prevents category errors.

It keeps grammar from becoming validation.

It keeps projection from becoming acceptance.

It keeps carriers from becoming identity.

It keeps implementation from becoming doctrine.

6. The 24-Cell Canon Spine

The current 24-file configuration is:

```
00-COSMOLOGY
01-ONTOLOGY
02-DOCTRINE
03-FOUNDATIONS
04-ATOMIC-KERNEL
05-PURE-ALGORITHMS
06-FORMAL-PROOFS
07-ISA
08-PROTOCOL-SEMANTICS
09-OMI-LISP
10-BLACKBOARD
11-NOTATION-MULTIPLEXING
12-VERSION-CONTROL
13-MASTER-ARCHITECTURE

14-CANONICAL-CONSTITUTION
15-CLASSIFICATION-TERMINOLOGY
16-SYMBOLIC-GRAMMAR
17-RELATIONAL-QUOTATION
18-GAUGE-INTERRUPTS
19-PROJECTION-FACES
20-TRANSFORM-CARRIERS
21-SURFACE-CLAUSES
22-PROPAGATION-MODEL
23-REFERENCE-GLOSSARY
24-OBSERVER-CHANGELOG
```

This should be read as a packed doctrinal lattice.

The first region constructs the system.

The second region classifies, quotes, projects, carries, propagates, and records the system.

A useful split is:

```
00-07 = formal origin and proof spine
08-12 = executable construction semantics
13     = master map
14-24 = constitutional, classificatory, projection, carrier, propagation,
reference, and observer layers
```

The master architecture does not sit above the canon.

It maps the canon.

The constitution does not execute the system.

It constrains authority.

The glossary does not create doctrine.

It stabilizes reference.

The changelog does not validate.

It records observer-visible change.

7. Relation to the Leech Lattice Mental Model

The Leech lattice is useful here as a mental model because it is a highly symmetric 24-dimensional lattice with exceptional packing properties.

OMI should not claim to be the Leech lattice.

Instead, OMI can borrow the discipline:

```
24-dimensional balance
high symmetry
no unnecessary roots
```

dense non-overlap
deep and shallow hole awareness

In OMI language:

24 dimensions of canon
balanced doctrinal cells
no arbitrary authority roots
dense relation coverage without overlap
awareness of gaps, holes, and unresolved declarations

A “hole” in this mental model is not a mathematical Leech hole inside OMI.

It is an architectural gap:

a declaration that almost fits,
but lacks a lawful neighbor,
witness,
closure,
or validation path.

Deep holes are major missing doctrines.

Shallow holes are local missing definitions.

The canon’s job is not to eliminate all future holes.

The canon’s job is to make holes visible and classifiable.

8. Relation to II25, 1 Mental Model

The Lorentzian lattice II25, 1 is useful as a mental model because it adds a time-like or direction-sensitive dimension to a highly structured 24-dimensional lattice picture.

For OMI, this suggests:

24 canon cells
+
direction of validation / propagation
=
bounded construction with lawful temporal orientation

Again, OMI should not claim literal equivalence.

The useful analogy is:

```
Leech-like 24-fold balance
+
Lorentzian directionality
=
canon plus validation direction
```

OMI has its own version of “time direction”:

```
Declaration
→ Omi-Notation
→ Omi-Citation
→ Validation
→ Accepted Omi-State
→ Omi-Attestation
→ Omi-Attribution
→ Omi-Projection
```

That direction must not be reversed.

Projection cannot validate.

Attestation cannot invent the relation.

OMI-Lisp cannot accept state.

A carrier cannot become authority.

This is OMI’s time-orientation rule.

9. The Omi-Ring as Local Tetrahedral Basis

The Omi-Ring supplies the local primitive basis.

The canon supplies the global bounded packing.

The Tetragrammatron adjudicates whether a candidate relation is lawfully placed.

The local tetrahedral basis can be stated as:

```
root
relation
witness
closure
```

or:

```
identity
quotation
adjudication
carry-forward
```

This is the smallest useful local cell.

A derived declaration must circumscribe this basis.

That means it must not merely say something.

It must be placeable as a relation.

It must be quotable without acceptance.

It must be witnessable.

It must have a closure path.

It must be adjudicable.

10. Karnaugh Maps as Adjudication Surface

A Karnaugh map is useful in OMI as a deterministic reduction surface.

It does not create truth.

It organizes candidate truth conditions so they can be reduced.

In OMI:

```
candidate declaration
→ truth gate
→ decision table
```

- Karnaugh reduction
- Tetragrammatron adjudication

The Karnaugh map is therefore not authority.

It is a reduction surface.

The Tetragrammatron adjudicates over the reduced relation.

The validation result determines whether the relation becomes accepted Omi-State.

Therefore:

Karnaugh maps reduce.
Tetragrammatron adjudicates.
Validation determines.
Attestation witnesses.
Projection displays.

11. Derived Declaration Rule

A derived declaration is valid only if it can be placed in the canon packing.

Placement requires:

1. It has a canonical root.
2. It quotes a relation rather than asserting authority.
3. It preserves place-value identity.
4. It declares its carrier face if carried.
5. It exposes a witness path.
6. It can be reduced by deterministic rules.
7. It can be adjudicated by the Tetragrammatron.
8. It can be recorded as accepted Omi-State only after validation.
9. It can be projected only after acceptance.

If any of these fail, the declaration remains unresolved.

Unresolved does not mean false.

Unresolved means:

not yet placed in the packing

12. Balance and Order

The reason to use Bravais and Leech lattice mental models is practical.

They teach:

do not place points arbitrarily
choose primitive basis carefully
preserve symmetry
track neighbors
avoid overlap
make holes visible
distinguish primitive cells from convenient cells
distinguish boundary from interior

That is exactly what the canon needs.

OMI should remain balanced by asking, for each proposed change:

Is this a primitive vector or a derived translation?
Is this a unit cell or a conventional convenience?
Is this a boundary document or an interior declaration?
Is this a relation, a projection, a carrier, or an authority claim?
Does it preserve the tetrahedral Omi-Ring basis?
Can the Tetragrammatron adjudicate it?

13. Canonical Lock

The 24-file OMI canon spine is the bounded n-sphere of OMI construction.

Each root document is a canonical boundary cell.

Together, the files form a sphere-packing configuration for relation construction.

The Omi-Ring supplies the local tetrahedral relation basis.

Derived declarations may circumscribe, refine, quote, or project within this packing, but they do not create new authority.

The Tetragrammatron adjudicates candidate declarations through deterministic reduction surfaces, including truth gates, decision tables, and Karnaugh maps.

Validation determines.

Omi-Attestation witnesses.

Accepted Omi-State may be recorded.

Projection only displays accepted relation state.

14. Short Form

The canon is not a list.

The canon is a packed boundary.

The Omi-Ring is the primitive local cell.

The 24-file spine is the global n-sphere.

Derived declarations are interior candidate balls.

The Tetragrammatron adjudicates placement.

Karnaugh maps reduce truth structure.

Validation determines.

Projection displays only accepted state.